Imagine that someone from Gotham city tries to order some French cheese: you look at your list of warehouses, compute the distance between your customer's mailing address and each of them, and you pick the closest one, P-5. Immediately after that, someone from Metropolis buys two wheels of the same cheese: unfortunately, you can't use any of the distances computed before, because the source point, the location of the customer, is completely different. So, you just go through that list of stores again, compute all the distances, and choose warehouse B-2. If the next request comes, say, from Civic City, off you go! You once again you need to compute all `N` distances again, to all your `N` warehouses.

8.2 Solutions

Now, I know, figure 8.1 only shows 5 warehouses, so it seems a trivial, quick operation, to go through all warehouses for each user. You could even handle orders manually, choosing case by case based on your guts feeling and experience.

But suppose that after one year, since your business is working well, more stores have decided to sell on your website, and you have something close to a hundred of them just in that same area. That's already challenging, and your customer care department can't cope with about a thousand orders a day: manually selecting the closest place for each order doesn't work anymore.